MIL-S-2410C
1 November 1978
SUPERSEDING
MIL-S-2410B
22 July 1960

MILITARY SPECIFICATION

SKETCHMASTER, OBLIQUE, 60-DEGREE,

PLOTTING SCALE RANGE 1/4 TO 1, WITH CHEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 60-degree depression angle oblique sketchmaster with chest. The sketchmaster is a means for transferring planimetric detail from 9- by 9-inch, 60-degree, oblique, aerial photographs to a base sheet or map manuscript. The sketchmaster has a plotting range of 0.25 to 1 times the isoline scale of the photography.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-B-613	- Brass, Leaded and Nonleaded, Flat Products (Plate, Bar, Sheet, and Strip).
QQ-C-390	- Copper, Alloy Castings (Including Cast Bar).
QQ-S-763	- Steel Bars, Shapes, and Forgings - Corrosion Resisting.
QQ-S-781	- Strapping, Steel, and Seals.
UU-T-101	- Tape, Gummed, Mending, and Reinforcing, (Paper and Cloth).
JJJ-C-561	- Cotton, Sterile (Purified) and Nonsterile.
NNN-P-40	- Paper, Lens.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-G-174	- Glass, Optical.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-12044	- Chests, Plywood.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

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DRAWINGS

ME

D7593-1 to 12 inclusive	- Sketchmaster, Oblique 60°, Plotting Scale 1/4 to 1.
TA13217E7448	- Chest, Sketchmaster, Oblique 60°.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The sketchmaster shall be as shown on Drawings D7593-1 to 12, inclusive, and as specified herein.

3.1.1 Chest. The chest shall be constructed in accordance with Top Assembly TA13217E7448 and MIL-C-12044.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the sketchmaster and chest. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced sketchmaster and chest). The contractor shall furnish one sketchmaster with chest for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce sketchmasters and chests that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Brass castings. Brass for castings shall conform to QQ-C-390, Alloy No. 852 and 854, Type I.

3.3.2 Brass sheet. Brass sheet shall conform to QQ-B-613, composition and temper best suited for the use.

3.3.3 Corrosion-resisting steel. Corrosion-resisting steel shall conform to QQ-S-763, Class 202.

3.4 Environmental conditions. The sketchmaster or chest shall not be damaged by storage under the following conditions:

Air temperature:

High - 155° F (68.3° C). Time period, 4 hours.
Low - Minus 50° F (-45.6° C). Time period, 6 hours. No solar radiation.

3.5 Mirrors. The reflecting surfaces of all mirrors shall be free from scratches, pin holes, or corrosive effects and shall show no signs of deterioration. The mirrors shall be of such optical flatness that no pair of interference fringes within the area of optical flat .250 inch (6.35 mm) from the edges shall be separated by less than .079 inch (2.0 mm). The mirrors shall be mounted on the instrument in such a manner as to eliminate strain, warp, or stress that would affect the flatness of the reflecting surfaces. In addition to the mirrors on the instrument, a spare mirror of each type shall be furnished with each instrument.

3.5.1 Fully reflecting mirror. The glass for the fully reflecting mirror shall conform to MIL-G-174, Grade A. All edges shall be ground. The mirror shall be front-surfaced with aluminum by evaporation so that the reflection of light in the visible spectrum shall be not less than 85 percent. The reflective layer shall be protected by a coating of silicon monoxide not less than 1,500 Angstrom units thick.

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3.5.2 Semitransparent mirror. The glass for the semitransparent mirror shall conform to MIL-G-174, Grade C. The reflective coating shall be nonabsorbing, all-dielectric film combination. A combination of film material having alternatively high and low refraction indices shall be used. The separate films of the coating shall be applied by vacuum evaporation to such thickness that the mirror shall provide a reflection of light in the visible spectrum of 70 percent, plus or minus 4 percent, and a transmission of light in the visible spectrum of 30 percent, plus or minus 4 percent (see 6.4). Absorption of visible light by the mirror, however, shall not exceed 4 percent.

3.6 Mirror protective covers. Protective covers for the mirrors on the instrument and replacement mirrors shall be as shown on the drawings. Cloth used for the covers shall be soft flannel which will not scratch or damage the mirrors.

3.7 Mirror cleaning material. One book or pad of 50 sheets of lens tissue paper conforming to NNN-P-40, Type I, Class 2, Size 4 by 6 inches, and a 1-ounce package of cotton conforming to JJJ-C-561, Grade A, shall be furnished with each sketchmaster.

3.8 Chest. The chest for the sketchmaster shall be as shown on TA13217E7448 and shall conform to the applicable requirements of MIL-C-12044. The following instructions, printed in bold black letters on a white background of plastic or metal, shall be fastened on the inside cover of each chest:

NOTICE

THE USE OF COTTON AND WATER OR A LIQUID CLEANER SUCH AS AMMONIA WATER IS RECOMMENDED FOR CLEANING THE MIRRORS OF THIS INSTRUMENT. USE EXTREME CAUTION TO PREVENT DAMAGE TO THE MIRRORS.

3.9 Fungus resistance. The chest and any part of the instrument (such as the web strap, mirror covers, and padding) which is subject to attack by fungus shall be of fungus-resistant materials or shall be treated to resist the growth of fungi. Fungus-resistant treatment shall not damage the mirrors.

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3.10 Treatment and painting. The chest and all parts of the instrument normally painted shall be treated and painted in accordance with MIL-T-704, Type A.

3.11 Identification marking. Each sketchmaster and its chest shall be identified in accordance with MIL-STD-130.

3.12 Workmanship. All parts and components of the sketchmaster and chests shall be free from burrs, sharp edges, and corners, except when sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced sketchmaster and chest shall be cause for disassembly, in the presence of a Government representative, of the first produced sketchmaster and chest to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced sketchmaster and chest. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced sketchmaster and chest inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced sketchmaster and chest inspection.

4.3.1 Examination. The first produced sketchmaster and chest shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first produced sketchmaster and chest shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sketchmaster and chest shall be examined as specified herein for the following major defects:

101. Dimensions of sketchmaster and chest not as shown on drawings.
102. Material not as specified.
103. Replacement mirror of each type not provided.

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104. Mirror edges not ground.
105. Reflecting surfaces of mirrors not protected as specified.
106. Components missing or do not fit properly.
107. Identification marking missing or illegible.
108. Workmanship not as specified.

4.5.2 Tests. Unless otherwise specified herein, the tests shall be conducted in the order listed:

4.5.2.1 Low-temperature storage. The sketchmaster and chest shall be subjected to a temperature of minus 50° F (minus 45.6° C) for not less than 6 hours. Upon completion of the low-temperature test, the instrument and chest shall be examined. Evidence of any change in shape or form of any metal parts, or deterioration of any wood, cloth fiber, reflective glass, or malfunction of any part or component of the sketchmaster and chest, peeling, blistering, discoloration or fading of any surface treatment or painting of any part or component caused by low temperature shall constitute failure of this test.

4.5.2.2 High-temperature storage. The sketchmaster and chest shall be subjected to a temperature of 155° F (68.3° C) for not less than 4 hours. Upon completion of the high-temperature test, the instrument and chest shall be examined. Evidence of any change in shape or form of any metal parts, or deterioration of any wood, cloth fiber, reflective glass, or malfunction of any part or component of the sketchmaster and chest, or signs of formation of fungi, peeling, blistering, discoloration or fading of any surface treatment or painting of any part or component caused by high temperature shall constitute failure of this test.

4.5.2.3 Flatness. An optical flat (certified by the National Bureau of Standards to be flat to within 1/4 fringe across the entire surface) with a diameter of not less than 6 inches and a semitransparent reflective coating on the flat side (the opposite face ground and polished) shall be placed on the surface to be tested, illuminated with light of 5,460 Angstrom units (Mercury), and adjusted until a minimum number of fringes are evident. Both contact surfaces shall be clean for this test. A standard scale graduated in millimeters shall be used to measure the perpendicular distance between interference fringes. Evidence of any pair of interference fringes separated by less than .079 inch (2 mm) shall constitute failure of this test.

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4.5.2.4 Reflection. Reflection shall be measured with a modified spectrometer or similar equipment which will maintain measurable angular relationships between a collimator (having an enclosed tungsten filament lamp, battery-powered at a color temperature of approximately 2,850° K), a mirror support, and a photocell which is corrected for eye sensitivity (equivalent to a Weston Photo Cell, Model 594, Type III, equipped with Viscor filter). The output of the photocell shall be measured with a laboratory microammeter having an accuracy of 0.5 percent of full reading or better. The collimator and photocell shall rotate in the same plane about a common center (like a spectroscope), and the mirror or reflector shall be mounted over the center of rotation and perpendicular to the plane of rotation of the collimator and photocell. Check the collimator light beam for infinity focus and uniform intensity. All measurements shall be made in a light-tight compartment or photographic dark room. To measure the reflectivity, orient the collimator and photocell 180 degrees apart (the mirror or reflector not being in place) and measure the output of the photocell. This is the reference or 100-percent reading. Mount the mirror or reflector over the center of rotation and perpendicular to the light beam. Orient the collimator until the light beam is incident on the mirror or reflector at an angle of 30 degrees measured from the normal to the reflecting surface. Orient the photocell in the reflected light beam until a maximum reading is obtained. The ratio of the two photocell measurements expressed as a percentage is the reflectivity of the mirror or reflector. Evidence of reflectance less than 85 percent shall constitute failure of this test.

4.5.2.5 Transmission, semitransparent mirror. To measure the transmissibility, obtain the initial photocell measurement as in the reflectivity measurement, then mount the reflector perpendicular to the light beam, leaving the photocell 180 degrees apart from the collimator. Record the output of the photocell. The ratio of the two photocell measurements expressed as a percentage is the transmissibility. The difference in the transmissibility and reflectivity of the reflector is its approximate absorption. Evidence of absorption of more than 4 percent shall constitute failure of this test.

4.5.2.6 Corrosion resistance. The mirror shall be inspected for scratches or pinholes with the mirror held between a 60-watt bulb and the eye. Evidence of scratches or pinholes shall constitute failure of this test. The mirror shall then be completely immersed in a

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6-percent sal solution (NaCl) at room temperature for not less than 12 hours. The mirror shall be removed from the solution, washed clean, and examined. Evidence of corrosive effects shall constitute failure of this test.

4.5.2.7 Abrasion resistance. A .750-inch (19.05 mm) steel ball covered with linen having a thread count of not more than 60 threads per inch shall be loaded with a 500-gram weight and then rubbed randomly on the reflecting surface with a reciprocating motion for not less than 100 cycles. The reflecting surface shall then be examined. Evidence of scratches or other signs of deterioration shall constitute failure of this test.

4.5.2.8 Coating adherence. Place the adhesive side of a length of tape conforming to UU-T-101, Type II, in contact with a portion of the coated surface and rub firmly against the surface, pulling the end of the tape down over the edge of the mirror. Slowly remove the tape and examine the mirror. Evidence of any removal of the coating from the mirror by the tape shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
110.	Box closure not Method V as specified.
111.	Strapping not as specified.
112.	Marking illegible, incorrect, incomplete, or missing.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each complete sketchmaster, replacement mirrors, mirror cleaning equipment, and technical publications contained in the chest shall be preserved in accordance with MIL-P-116, Method IIb. The chest shall be the inner container and the outer container shall be a close-fitting box conforming to PPP-B-636, W5c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Each complete sketchmaster, with chest, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Sketchmasters, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Sketchmasters shall be packed as specified in 5.2.1 for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. Sketchmasters, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The sketchmaster and chest is used in compilation of planimetric detail from 9- inch 9-inch oblique photographs.

6.2 Ordering data. Procurement documents should specify the followihg:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first produced sketchmaster and chest (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When strapping shall be Finish A (see 5.2.1).

6.3 First produced sketchmaster and chest. Any changes or deviations of production sketchmasters and chests from the approved first produced sketchmaster and chest during production will be subject to the approval of the contracting officer. Approval of the first produced sketchmaster and chest will not relieve the contractor of his obligation to furnish sketchmasters and chests conforming to this specification.

6.4 Semitransparent mirror. A combination of film materials having alternatively high and low refraction indices must be used to obtain the high reflecting coating for the semitransparent mirror. A coating of magnesium fluoride, silicon monoxide, magnesium fluoride, titanium dioxide (rutile) films, each 1/4 wave length in thickness deposited on the glass in that order will meet specification requirements (see 3.5.2).

6.5 Data requirements. The contracting officer should include requirements for data such as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each sketchmaster and chest.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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DOCUMENT IDENTIFIER AND TITLE MIL-S-2410C Sketchmaster, Oblique, 60-Degree, Plotting Scale Range 1/4 to 1, with Chest

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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